F29H85x MCAL 01.02.00 Release Notes

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Introduction

F29H85x package contains MCAL and CDD modules as per AUTOSAR architecture. These software modules are developed specific for platform hardware to enable the direct access to on-chip MCU peripherals. Thus making the upper layer of AUTOSAR architecture independent of MCU platform hardware. These drivers are developed as per AUTOSAR version 4.3.1. The delivery contains source code of all drivers, EB Tresos plugin to generate configuration data, bare-metal examples of each driver and its user guide.

F29H85x MCAL v01.02.00 is an early adopter release and not functional safety compliant targeted for F29H85x.

Device Support

The following devices are supported with this release

- MCU's
  - F29H859TU-Q1

- Hardware Platforms
  - F29H85X-SOM-EVM

What Is Supported

- MCAL / CDD (AUTOSAR **v4.3.1**):

Drivers	**Modules**
Micro controller Drivers	GPT, MCU, WDG
Communication Driver	CAN, LIN, SPI
I/O Drivers	DIO, PORT
Memory Drivers	FLS
Complex Device Drivers	ADC, XBAR, SENT, IPC, UART, PWM, ECAP

- Reference Code:
  - Multi-Core startup code
- Build System:
  - CMake
  - CCS Projectspec

Note: The minimum version of CCS required is CCSv20.01.01 and compiler version is C29Clang 1.0.0 LTS. Please refer to the Dependencies section for the download links.

What Is Not Supported

- Multi-core MCALs as per AUTOSAR 4.3.1.

New In This Release

- MCAL / CDD :
  - Newly added modules -
    - CDD ECAP
    - CDD UART
    - CDD PWM
    - CDD IPC
      - PduR Integration
    - FLS
  - Existing modules with new functionality -

- CDD SENT
  - MTP Mode
  - PduR Integration
- MCU, PORT, DIO, GPT, CAN, WDG, SPI, LIN
  - Safety Mechanisms are enabled
- CDD ADC and CDD XBAR
  - Autosar compliant updated Schema

- Reference Code: Multi-Core Startup

Module Version Number

MCAL / CDD	Module Version Number
MCU	01.01.00
PORT	01.01.00
DIO	01.00.02
GPT	01.00.02
CAN	01.01.00
WDG	01.00.01
LIN	01.01.00
SPI	01.01.00
FLS	01.00.00
CDD ADC	02.00.00
CDD XBAR	02.00.00
CDD SENT	02.00.00
CDD PWM	01.00.00
CDD ECAP	01.00.00
CDD IPC	01.00.00
CDD UART	01.00.00

Note:

We follow Autosar's requirement [SRS_BSW_00321] for versionoing of each module-

5.2.4.10 [SRS_BSW_00321] The version numbers of AUTOSAR Basic Software Modules shall be enumerated according specific rules

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Type:	Valid
Description:	The version numbers of AUTOSAR Basic Software Modules shall be enumerated according to the following rules: - Increasing a more significant digit of a version number resets all less significant digits - The PATCH_VERSION is incremented if the module is still upwards and downwards compatible (e.g. bug fixed) - The MINOR_VERSION is incremented if the module is still downwards compatible (e.g. validfunctionality added) - The MAJOR_VERSION is incremented if the module is not compatible any more (e.g. existing API valid)
Rationale:	Provide unambiguous version identification for each module, provide version cross check as well as basic version retrieval facilities. Compatibility is always visible!
Use Case:	Example: ADC module with version 1.14.2:

Fixed In This Release

Key	Summary	Component/s	Severity
MCAL-29915	MCAL CAN Module Interrupt Configuration issue	Can	S2-Major
MCAL-29487	The parameter passed to the CanIf_TxConfirmation function upon completion of MCAL CAN transmission is incorrect.	Can	S2-Major
MCAL-28922	Logical errors in Can Deinit and Icom Handling	Can	S3-Minor
MCAL-27721	Lin High Timeout Value for Go To Sleep	Lin	S3-Minor
MCAL-27717	Incorrect handling of Wdg initial and maximum timeout configuration parameters	Wdg	S3-Minor
MCAL-26930	Can Wake-up is not working on F29H85x	Can	S3-Minor

6 issues

Known Issues And Limitations

There are few limitations and known issues associated with this release

Limitations

- CDD PWM - Following sub-modules are not supported:

- Dead-band Submodule
- Chopper Submodule
- Trip-Zone Submodule
- Diode Emulation Submodule
- MINDB & ICL submodules
- Event-trigger Submodule
- Digital Compare Submodule

- CDD ECAP -
  - Cdd Ecap driver doesn't support APWM mode (single channel PWM generation from ECAP)
  - Cdd Ecap driver doesn't support ECAP Synchronization (SYNCIN)
  - Cdd Ecap doesn't support Signal monitoring units
  - Cdd Ecap doesn't operate ECAP in Single-shot capture of up to four event time-stamps

- Test Infrastructure : Following tests have not been performed due to limitations of available test infrastructure -
  - CDD SENT : Following modes in SENT are not tested -
    - Enhanced 16-bit Slow Mode.
    - Nibble Count 1,2,3,4,5.
    - MTP Mode with all 4 sensors connected.
  - CDD IPC : IPC is validated between CPU1 and CPU3. Since, CPU2 is configured in lockstep in our framework; IPC between CPU1 and CPU3 or CPU2 and CPU3 is not validated.
  - Fls : Timeout tests are not performed as FLASH characterization data is not yet available for this device.
  - Spi : Board to board Spi communication validated upto 40MHz. 50MHz have been only test in loopback mode.

New Known Issues

Key	Summary	Component/s	Workaround	Severity
MCAL-30268	MainFunction Write schedulable entity and timing event is not generated in Cdd_Uart_BSWMD.arxml	CDD UART	Add Cdd_Uart_MainFunctionWrite manually and map it to timing event as per requirement of the project	S3-Minor
MCAL-30149	CDD ECAP: First rising edge not detected/not counted in BOTH_EDGE activation condition	CDD ECAP	None. When activation condition is configured as both edges (in edge detect or edge count measurement mode), provide a falling edge as the first edge in the input to ECAP	S3-Minor
MCAL-30035	Mixed mode handling is incorrect in Can	Can	The processing mode for transmit/receive message can be configured to either polling or interrupt for Can controller.	S3-Minor

3 issues

Existing Known Issues

Key	Summary	Component/s	Workaround	Severity

No issues found

Dependencies

This release has dependency on the following tools

- CCS v20.01.01.
- C29Clang Compiler 1.0.0 LTS.
- EB Tresos (ACG 8.8.10) : Request access here.

References

1. F29H85x MCAL User Guide
2. F29H85x Technical Reference Manual and Device Datasheet : F29H859TU-Q1.

Technical Support

C2000 microcontroller e2e forum